Myopathy, Congenital, with Tremor (CMYO16 / "Myotrem") — Comprehensive Disease Report

Target disease: Myopathy, Congenital, with Tremor **MONDO ID:** MONDO:0032797 · **OMIM:** #618524 · **MedGen:** C5231401 (UID 1684886) **Causal gene: MYBPC1** (slow skeletal myosin-binding protein C; sMyBP-C) — HGNC:7549; NCBI Gene 4604; UniProt Q00872; chromosome 12q23.2

Summary

Myopathy, Congenital, with Tremor (MONDO:0032797) is Congenital Myopathy-16 (CMYO16), also called "Myotrem" or myogenic-tremor myopathy (OMIM #618524). Its causal gene is *MYBPC1*, which encodes slow skeletal muscle myosin-binding protein C (sMyBP-C). This is an autosomal dominant skeletal-muscle disorder characterized by onset in infancy of hypotonia and a distinctive high-frequency (~10–11 Hz) tremor of the limbs, together with mild proximal and axial muscle weakness. The clinical course is comparatively benign: weakness tends to stabilize or improve after adolescence, and the condition is managed supportively.

An important note for knowledge-base curation: the disease name "Myopathy, Congenital, with Tremor" is easily confused with a separate, single de novo case of a tremor-associated myopathy caused by *MYH7* (β-cardiac/slow skeletal myosin heavy chain) reported in 2023. During this investigation an early draft mis-assigned the gene to *MYH7*; that was corrected. **The MYH7 tremor case is explicitly described by its own authors as a "first-of-its-kind" report distinct from the classical MYBPC1 entity.** Following the MONDO/OMIM identifiers given in the research template, the disease in scope here is unambiguously the *MYBPC1* disease, CMYO16. *MYH7* findings are retained in this report only as clearly-labeled comparative/contextual material.

Mechanistically, all reported pathogenic variants are heterozygous missense changes clustered in the N-terminal **M-motif** of sMyBP-C (e.g., p.Tyr247His, p.Glu248Lys, p.Leu259Pro, p.Leu263Arg). These variants dysregulate the binding of sMyBP-C's N-terminus to myosin and perturb actomyosin cross-bridge cycling and stretch activation in slow (type-1) muscle fibers. The resulting aberrant, pulsatile myofilament activity is the proposed sarcomeric origin of the "myogenic tremor." The disease is ultra-rare (~16 patients from ~8 families reported by mid-2024), has a favorable natural history, and is modeled by an E248K knock-in mouse that recapitulates the myopathy.

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1. Disease Information

Overview. CMYO16 is an autosomal dominant congenital myopathy defined by the combination of infantile hypotonia, mild and comparatively non-progressive skeletal muscle weakness (proximal and axial predominant), and a hallmark high-frequency postural/action tremor of myogenic (sarcomeric) origin. OMIM/MedGen summarize it as: *"Congenital myopathy-16 (CMYO16) is an autosomal dominant muscle disorder characterized by onset of hypotonia and tremor in infancy... proximal muscle weakness, and a high-frequency tremor of the limbs... course tends to stabilize after adolescence (summary by Stavusis et al., 2019)."*

Key identifiers.

Resource	Identifier
MONDO	MONDO:0032797
OMIM (phenotype)	#618524 (CMYO16)
OMIM (gene)	*160794* (MYBPC1)
MedGen	C5231401 / UID 1684886
HGNC	7549 (MYBPC1)
NCBI Gene	4604
UniProt	Q00872 (MYPC1_HUMAN)
Orphanet	No dedicated code identified; falls under "congenital myopathy" grouping
ICD-11	8C70 (Congenital myopathies) — closest mapping
ICD-10	G71.2 (Congenital myopathies) — closest mapping

Synonyms / alternative names: Congenital Myopathy-16 (CMYO16); "Myotrem"; Myogenic tremor myopathy; Myopathy, congenital, with tremor.

Information source type: Aggregated disease-level resources (OMIM, MedGen, HPO) combined with a small number of primary case series (individual-patient clinical/genetic data from ~8 families). There is no large EHR/registry dataset for this ultra-rare disorder.

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2. Etiology

Primary cause — genetic. CMYO16 is a monogenic Mendelian disorder caused by heterozygous (autosomal dominant) missense variants in *MYBPC1*. There is no known environmental, infectious, or toxic cause. The disease arises either from inheritance within multigeneration families or from de novo variants.

Genetic risk factors. The causal variants themselves are the risk determinant. They cluster in the NH2-terminal **M-motif** of sMyBP-C: - **p.Tyr247His (Y247H)** and **p.Glu248Lys (E248K)** — segregating in two three-generation families (Stavusis et al., 2019; PMID: 31025394). - **p.Leu263Arg (L263R)** — de novo in 3 individuals; **p.Leu259Pro (L259P)** — de novo in 1 individual (Shashi et al., 2019; PMID: 31264822).

These variants are absent from population controls, affect residues conserved across vertebrates, and are predicted damaging (ACMG pathogenic/likely pathogenic; recurrent de novo events support pathogenicity).

Environmental risk factors / protective factors / gene–environment interactions: None identified. No modifier alleles, protective variants, dietary, lifestyle, occupational, or infectious factors have been reported. Given the Mendelian, highly penetrant nature of the condition, environmental contributions are not expected to be major.

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3. Phenotypes

The phenotype spectrum and official HPO frequencies are drawn from the 7-patient cohort annotated to OMIM:618524 (Stavusis et al., 2019; PMID: 31025394).

Phenotype	HPO term	Frequency	Type
Flexion contracture	HP:0001371	7/7 (100%)	Physical manifestation
Postural tremor	HP:0002174	6/7 (86%)	Clinical sign
Axial muscle weakness	HP:0003327	6/7 (86%)	Clinical sign
Scoliosis	HP:0002650	5/7 (71%)	Physical manifestation
Tongue tremor	HP:0031947	3/7 (43%)	Clinical sign
Scapular winging	HP:0003691	3/7	Clinical sign
Spinal rigidity	HP:0003306	3/7	Clinical sign
Lumbar hyperlordosis	HP:0002938	3/7	Physical manifestation
High palate	HP:0000218	3/7	Physical manifestation
Proximal muscle weakness	HP:0003701	present	Clinical sign
Distal muscle weakness	HP:0002460	present	Clinical sign
Myopathic EMG abnormalities	HP:0003458	2/2	Laboratory abnormality
Micrognathia	HP:0000347	minor	Physical manifestation
Narrow mouth	HP:0000160	minor	Physical manifestation
Prominent nasolabial fold	HP:0005272	minor	Physical manifestation
Autosomal dominant inheritance	HP:0000006	—	Inheritance

Additional recognized features reported in later series include neonatal respiratory distress with **laryngeal stridor** and **undulating/undulatory tongue movements** (Neurol Clin Pract 2024, PMID: 38690148; Pediatr Neurol 2023, PMID: 37392669).

Phenotype characteristics. - **Age of onset:** Infancy/congenital — hypotonia and tremor present from early infancy. - **Severity:** Mild overall; tremor is the most conspicuous and functionally relevant feature. - **Progression:** Weakness is mild and tends to stabilize or improve after adolescence (gradual clinical improvement). Tremor persists. Scoliosis and contractures may require orthopedic attention. - **Frequency:** See table above (per-phenotype percentages from the core cohort).

Quality-of-life impact: The persistent high-frequency tremor can impair fine motor tasks and handwriting; scoliosis, flexion contractures and spinal rigidity may affect posture, mobility, and comfort. Formal QoL instrument data (EQ-5D, SF-36, PROMIS) are not available for this ultra-rare disease. Overall, because weakness is mild and tends to improve, long-term functional prognosis is relatively favorable.

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4. Genetic / Molecular Information

Causal gene. *MYBPC1* (OMIM *160794; HGNC:7549; NCBI Gene 4604; UniProt Q00872), chromosome 12q23.2, encoding slow skeletal muscle myosin-binding protein C (sMyBP-C)*. sMyBP-C is a thick-filament accessory protein of the C-zone that regulates actomyosin interactions and contractile kinetics; it also functionally recruits muscle creatine kinase to the myosin filament (NCBI Gene 4604 summary).

Pathogenic variants (CMYO16-causing subset). All are heterozygous **missense** changes in the **N-terminal M-motif**:

Variant (protein)	Domain	Families / individuals	Origin	Reference
p.Tyr247His (Y247H)	M-motif	one 3-generation family	inherited (AD)	Stavusis 2019 (PMID: 31025394)
p.Glu248Lys (E248K)	M-motif	one 3-generation family	inherited (AD)	Stavusis 2019 (PMID: 31025394)
p.Leu259Pro (L259P)	M-motif	1 individual	de novo	Shashi 2019 (PMID: 31264822)
p.Leu263Arg (L263R)	M-motif	3 individuals	de novo	Shashi 2019 (PMID: 31264822)

Variant classification: Pathogenic / likely pathogenic by ACMG/AMP criteria (absent from controls, conserved residues, recurrent de novo, functional evidence). **Variant type/class:** Missense (single-nucleotide substitutions). **Allele frequency:** Absent or extremely rare in population databases (gnomAD); not present in healthy controls. **Somatic vs germline:** Germline (inherited or de novo germline). **Functional consequences:** Not simple loss of function. Biochemically, **Y247H and E248K markedly INCREASE N-terminal binding of**

sMyBP-C to myosin, whereas **L263R DECREASES M-motif binding to myosin** — divergent biochemical effects that converge on dysregulated actomyosin cross-bridge cycling. This is best described as a dominant gain-of-dysfunction / dysregulatory mechanism rather than haploinsufficiency.

Allelic disorders (same gene, different phenotype — important for ClinVar interpretation). ClinVar lists ~52 pathogenic/likely-pathogenic *MYBPC1* records (of ~523 total), but only the small set of N-terminal M-motif missense variants cause CMYO16. Other *MYBPC1* variants cause: - **Distal arthrogryposis type 1B (DA1B)** — OMIM 614335. - **Lethal congenital contracture syndrome 4 (LCCS4)** — OMIM 614915 (recessive, lethal).

Modifier genes / epigenetics / chromosomal abnormalities: None reported. No DNA-methylation, histone, or large-scale chromosomal abnormality is implicated. Sex-dependent modification of progression is suggested by the mouse model (below) but no human modifier gene is defined.

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5. Environmental Information

No environmental, lifestyle, or infectious factors are implicated in CMYO16. It is a purely genetic Mendelian disorder. There are no reported toxic, occupational, dietary, or pathogen-related contributors. **(Not applicable.)**

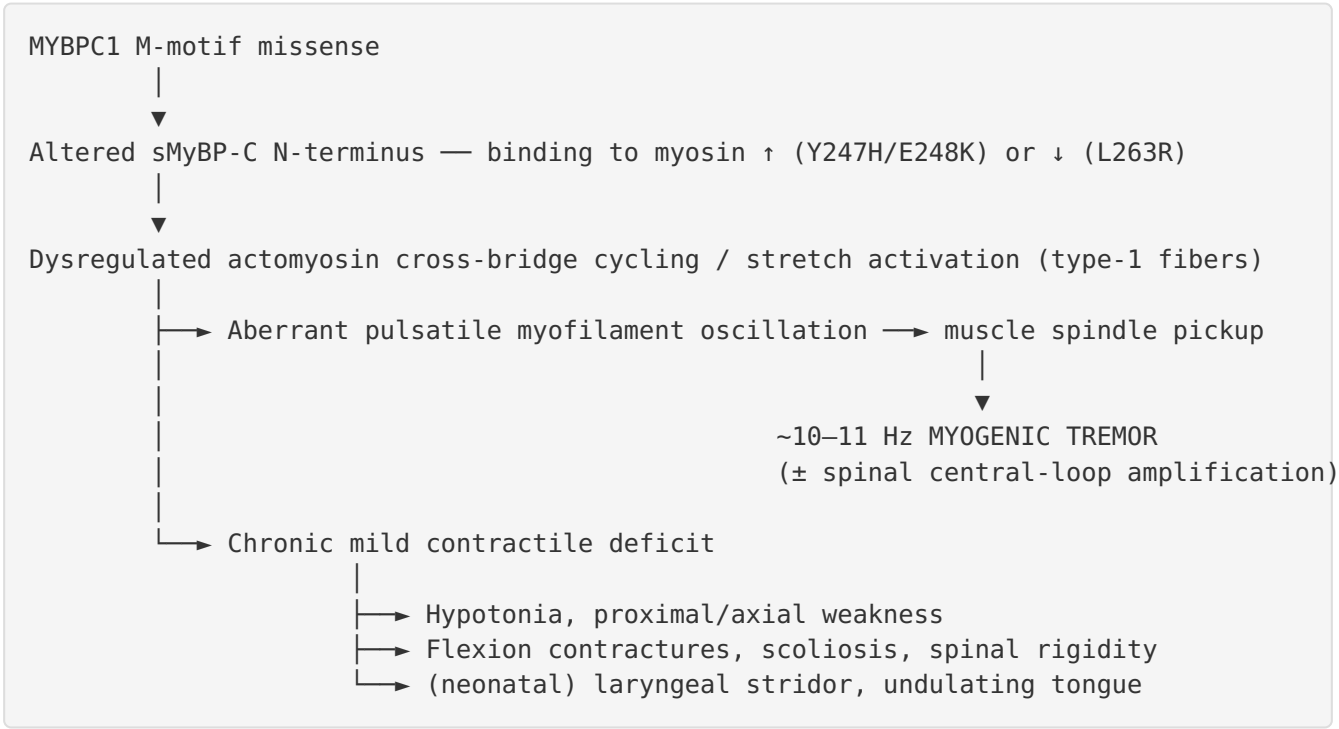
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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A heterozygous **missense variant in the N-terminal M-motif of *MYBPC1*** (e.g., Y247H, E248K, L259P, L263R) **results in** an altered sMyBP-C N-terminus.
2. The altered M-motif **changes the strength of sMyBP-C's binding to the myosin filament** — increased binding (Y247H/E248K) or decreased binding (L263R). *(Demonstrated biochemically; Stavusis 2019, Shashi 2019.)*
3. Altered sMyBP-C–myosin tethering **dysregulates actomyosin cross-bridge cycling and stretch activation** within the C-zone of slow (type-1) fibers. *(Inferred from biochemistry + myofilament physiology.)*

4. This **leads to** aberrant, pulsatile/oscillatory myofilament activity in slow-twitch fibers (aberrant stretch activation). *(Supported by ex-vivo myofilament studies; Int J Mol Sci 2025, PMID: 40508067.)*
5. Sarcomere-level oscillation **is picked up by muscle spindles** and **produces** a high-frequency (~10–11 Hz) postural/action tremor — the **"myogenic tremor"** — potentially amplified by a segmental spinal central loop. *(Proposed mechanism; sarcomere-origin model supported by electrophysiology in the comparative MYH7 case, PMID: 37070061.)*
6. In parallel (branch), chronic contractile dysfunction of slow fibers **results in** mild hypotonia, proximal/axial weakness, and secondary musculoskeletal features (flexion contractures, scoliosis, spinal rigidity).
7. In some neonates, involvement of laryngeal/pharyngeal musculature **leads to** stridor and respiratory distress and undulating tongue movements.



Detail by category

- **Molecular pathways / biochemical abnormalities:** The core defect is in the **sarcomeric thick-filament regulatory system**. sMyBP-C normally modulates the kinetics of myosin cross-bridge attachment via its N-terminal M-motif interacting with myosin S2/head and actin. Variants shift this equilibrium, altering cross-bridge cycling and stretch-activation responses. No canonical signaling cascade (Wnt/MAPK/mTOR/PI3K-AKT) is implicated — this is a structural/mechanochemical sarcomere disease.

- **Cellular processes:** Dysregulated excitation–contraction coupling and myofilament mechanics in slow-twitch skeletal myofibers; aberrant pulsatile activity. No prominent apoptosis, autophagy, or inflammation is reported as primary.
- **Protein dysfunction:** Dominant gain-of-dysfunction. Both increased and decreased sMyBP-C–myosin binding are pathogenic, indicating that precise tuning of this interaction is required and that perturbation in either direction disrupts contractile regulation.
- **Metabolic changes:** sMyBP-C recruits muscle creatine kinase (CKM) to the myosin filament, linking the contractile machinery to local ATP buffering; perturbation may locally affect energy delivery to the cross-bridge, though this is not fully established in CMYO16.
- **Immune involvement:** None.
- **Tissue damage mechanisms:** Primarily functional/mechanical rather than degenerative; muscle biopsy shows myopathic changes (see Diagnostics).
- **Suggested GO terms:** GO:0006936 (muscle contraction), GO:0030049 (muscle filament sliding), GO:0032036 (myosin heavy chain binding), GO:0005859 (muscle myosin complex), GO:0030017 (sarcomere, cellular component).
- **Suggested CL terms:** CL:0000189 (slow muscle cell), CL:0008002 (skeletal muscle fiber), CL:0000188 (cell of skeletal muscle).

Molecular profiling

No transcriptomic, proteomic, metabolomic, or lipidomic disease-signature datasets are published for CMYO16 specifically. Mechanistic evidence comes from targeted biochemistry (binding assays), ex-vivo myofilament physiology, and the knock-in mouse (below).

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7. Anatomical Structures Affected

- **Organ/body system level:** Skeletal muscle (musculoskeletal system). Secondary involvement of the axial skeleton (scoliosis, spinal rigidity, lumbar hyperlordosis) and, in neonates, the **larynx/upper airway** (stridor) and **tongue** (tremor/undulating movements).
- **Tissue level:** Striated skeletal **muscle tissue**, with predominant effect on **slow (type-1) fibers** (sMyBP-C is the slow-fiber MyBP-C isoform). UBERON:0001134 (skeletal muscle tissue); UBERON:0003672 (axial skeletal muscle); UBERON:0001737 (larynx); UBERON:0001723 (tongue).

- **Cell level:** Slow skeletal muscle fibers (CL:0000189); muscle spindle sensory apparatus is implicated in tremor propagation.
- **Subcellular level:** The **sarcomere** (GO:0030017), specifically the **C-zone of the A-band thick filament** where MyBP-C resides; the **muscle myosin complex** (GO:0005859). No mitochondrial, nuclear, ER, or lysosomal primary defect.
- **Localization / lateralization:** Bilateral and generally symmetric; proximal and axial predominance of weakness; tremor affects limbs (postural/action) and can involve facial/tongue musculature.

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8. Temporal Development

- **Onset:** Congenital/infantile — hypotonia and tremor from early infancy; onset pattern is insidious/chronic (not acute).
- **Progression:** Mild and slowly evolving; **weakness tends to stabilize or improve after adolescence** ("gradual clinical improvement"). Tremor typically persists. Scoliosis/contractures may progress and need monitoring.
- **Course pattern:** Chronic, lifelong, but non-degenerative and relatively stable/improving — distinguishing it from progressive muscular dystrophies.
- **Critical periods:** The neonatal period carries the main acute risk (respiratory distress/stridor requiring non-invasive ventilation in a minority). Childhood/adolescence is the window of functional improvement.

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9. Inheritance and Population

- **Inheritance: Autosomal dominant** (HP:0000006). Both inherited (multigeneration families) and de novo variants reported.
- **Penetrance:** Appears high in reported families (variants segregate with disease across three generations), though the small sample precludes precise estimates.
- **Expressivity:** Variable — ranging from predominantly tremor with minimal weakness to neonatal respiratory compromise.

- **Mosaicism / anticipation / founder effects / consanguinity:** No genetic anticipation (not a repeat-expansion disorder). No founder effect or consanguinity role (dominant; de novo events documented). Germline mosaicism not specifically reported for human CMYO16 (notably documented in the porcine MYH7 model — a different gene — see comparative section).
- **Carrier frequency:** Not applicable in the recessive sense (dominant disorder); causal variants are vanishingly rare in the population.
- **Epidemiology: Ultra-rare.** By mid-2024, approximately **16 patients from ~8 families** had been reported (13 patients/6 families plus 3 new patients/2 families; Neurol Clin Pract 2024, PMID: 38690148). No prevalence/incidence estimates per 100,000 are available. No sex predilection established in humans (the mouse model shows sex-dependent progression). No specific geographic/ethnic clustering reported.

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10. Diagnostics

Clinical / laboratory: - **EMG:** Myopathic abnormalities (HP:0003458), present in those tested. - **Serum CK:** Typically normal or mildly elevated (congenital myopathy pattern). - **Muscle biopsy:** Myopathic changes; specific uniform CMYO16 histopathology is not firmly codified given the small cohort. - **Tremor characterization:** Quantitative accelerometry/EMG shows a distinctive **high-frequency ~10–11 Hz postural/action tremor**; this myogenic signature is a key diagnostic clue distinguishing it from central (e.g., essential, parkinsonian) tremor. - **Imaging:** Muscle MRI and spine imaging (for scoliosis) are supportive but no pathognomonic pattern is defined.

Genetic testing (definitive): - **First-line:** Gene panel for congenital myopathies including *MYBPC1*, or exome/genome sequencing. - **Targeted/single-gene:** *MYBPC1* sequencing, focusing interpretation on **N-terminal M-motif missense variants** (CMYO16), while recognizing that other *MYBPC1* variants cause DA1B (OMIM 614335) or LCCS4 (OMIM 614915). - WES/WGS are highly useful given phenotypic overlap with other congenital myopathies. CMA, karyotyping, FISH, mtDNA testing, and repeat-expansion testing are **not** indicated (no structural, mitochondrial, or repeat basis).

Clinical criteria / differential diagnosis: No formal consensus diagnostic criteria exist. Differential diagnoses include other congenital myopathies (nemaline, core, centronuclear), other MyBP-C/arthrogryposis phenotypes, central tremor disorders, and the rare *MYH7*

tremor-myopathy case. The combination of congenital hypotonia + mild non-progressive weakness + a sarcomere-origin high-frequency tremor + an M-motif *MYBPC1* variant is diagnostic.

Screening: No population newborn screening. Cascade/predictive genetic testing and genetic counseling are appropriate for at-risk relatives in known families.

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11. Outcome / Prognosis

- **Survival/mortality:** Generally favorable; the disorder is not typically life-limiting. The main acute risk is **neonatal respiratory impairment** (laryngeal stridor requiring non-invasive ventilation) in a minority of cases.
- **Morbidity/function:** Mild myopathy "invariably associated with a distinctive myogenic tremor and hypotonia with gradual clinical improvement" (Neurol Clin Pract 2024, PMID: 38690148). Persistent tremor and musculoskeletal features (scoliosis, contractures) are the main long-term morbidities.
- **Recovery potential:** Weakness tends to stabilize/improve after adolescence; tremor persists.
- **Prognostic factors:** Neonatal respiratory involvement marks a more severe early course; otherwise prognosis is good.
- **QoL measures:** No disease-specific validated instruments reported.

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12. Treatment

No disease-modifying or gene-specific therapy exists. Management is **supportive and multidisciplinary**: - **Respiratory:** Neonatal monitoring; non-invasive ventilation for stridor/respiratory distress when present. - **Orthopedic:** Monitoring and management of scoliosis, flexion contractures, spinal rigidity (bracing, physiotherapy, surgery if indicated). - **Rehabilitation:** Physical therapy, occupational therapy; stretching for contractures; mobility support. - **Tremor management:** No established pharmacotherapy proven for this myogenic tremor; because it is sarcomere-origin rather than central, classic central-tremor drugs may be less effective (unproven). - **Pharmacogenomics / gene therapy / cell therapy / RNA therapy / immunotherapy:** None available or in trials specifically for CMYO16. - **Experimental:** No

registered disease-specific clinical trials (ClinicalTrials.gov) identified. Future sarcomere-modulating agents are conceptually relevant but untested here. - **Genetic counseling** is a core component (autosomal dominant; 50% transmission risk; de novo recurrence counseling).

Suggested NCIT intervention concepts: Supportive Care; Physical Therapy; Occupational Therapy; Mechanical Ventilation; Orthopedic Surgery; Genetic Counseling.

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13. Prevention

- **Primary prevention:** Not applicable (genetic disorder). **Genetic counseling** and, where desired, **preimplantation genetic testing (PGT)** or **prenatal testing** in families with a known variant.
- **Secondary prevention:** Early detection of at-risk relatives via cascade genetic testing; early neonatal respiratory surveillance in affected pedigrees.
- **Tertiary prevention:** Proactive orthopedic and respiratory management to prevent complications (scoliosis progression, contracture, airway compromise).
- **Immunization / public health / environmental / prophylaxis:** Not applicable.

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14. Other Species / Natural Disease

- **Human gene / orthologs:** *MYBPC1* (NCBI Gene 4604). Orthologs exist across vertebrates (the affected M-motif residues are conserved across vertebrates, which underpins their pathogenicity).
- **Natural disease in other species (same gene):** No well-characterized naturally occurring *MYBPC1* M-motif tremor-myopathy is documented in companion animals or livestock to date.
- **Comparative note (different gene, analogous phenotype):** A naturally occurring **MYH7** tremor-myopathy, "**Campus syndrome (CPS),**" occurs in pigs — a de novo germline (and somatic mosaic) in-frame insertion in *MYH7* exon 30 (c.4320_4321insCCCGCC; p.Ala1440_Ala1441insProAla) causing progressive postural tremor in ~9% of a founder boar's offspring, with dominant inheritance confirmed by backcross (PMID: 23153285). This is a *different gene* from CMYO16 but illustrates that sarcomere thick-filament defects can

produce dominant tremor-myopathy across species. It is included only for comparative biology; it is **not** the MONDO:0032797 disease.

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15. Model Organisms

- **Mouse (primary model):** An **E248K knock-in mouse (E248K-KI)** carrying the human variant recapitulates the myopathy and shows **sex-dependent, muscle-specific progression with aging** (Dominguez-Martin / Kontrogianni-Konstantopoulos group, *JCI Insight* 2025;10(15):e182471, PMID: 40569690). This is the key tool for studying disease mechanism and testing interventions.
- **Ex-vivo myofilament studies:** Implicate **aberrant stretch activation / pulsatile myofilament activity** in slow-twitch fibers as a tremor driver (*Int J Mol Sci* 2025;26:5252, PMID: 40508067).
- **Model strengths:** The E248K-KI reproduces the myopathy and allows dissection of age- and sex-dependent progression and myofilament physiology.
- **Model limitations:** Tremor phenotyping in mice is challenging; central-loop amplification of a myogenic tremor may differ between species; only one variant (E248K) is modeled as a knock-in so far.
- **Resources:** MGI (mouse *Mybpc1*); model generated by academic groups rather than large repositories.

Mechanistic Model / Interpretation

CMYO16 is best understood as a **sarcomere thick-filament regulatory disease** in which precise tuning of the sMyBP-C N-terminal M-motif's interaction with myosin is essential for normal slow-fiber contractile kinetics. The striking biochemical observation that **opposite effects on binding (increased for Y247H/E248K, decreased for L263R) both cause the same clinical entity** indicates that the M-motif functions as a finely balanced mechanochemical regulator; perturbation in either direction destabilizes cross-bridge cycling and stretch activation. The resulting aberrant, oscillatory myofilament activity in slow-twitch fibers is

sensed by muscle spindles and manifests as a distinctive ~10–11 Hz **myogenic tremor** — a tremor whose origin is the muscle itself rather than the central nervous system, possibly reinforced by a segmental spinal reflex loop.

This "bottom-up" tremor model is conceptually supported by electrophysiological work on an analogous sarcomere-myopathy patient, in whom the authors proposed that *"the tremor originates at the sarcomere level within muscles, which is then picked up by muscle spindles and leads to activating input to the neuraxis segment"* (PMID: 37070061). The branch to weakness/hypotonia reflects chronic mild contractile inefficiency, and the musculoskeletal and airway features are downstream mechanical consequences. The E248K knock-in mouse and ex-vivo myofilament studies provide experimental validation of the sarcomeric-origin hypothesis.

Evidence Base

PMID	Role in this report	Key support
31025394 (Stavusis 2019, *Ann Neurol*)	Foundational	Identified Y247H/E248K in two 3-generation families; defined "myogenic tremor"; source of OMIM:618524 HPO frequencies
31264822 (Shashi 2019, *Hum Mutat*)	Foundational	De novo L263R (×3) and L259P (×1); divergent M-motif binding effects
38690148 (Neurol Clin Pract 2024)	Epidemiology/ natural history	~16 patients/~8 families; mild myopathy with tremor + gradual improvement; neonatal stridor
37392669 (Pediatr Neurol 2023)	Phenotype	Respiratory distress with stridor; undulating tongue movements
40569690 (*JCI Insight* 2025)	Model organism	E248K knock-in mouse recapitulates myopathy; sex-/ muscle-specific progression
40508067 (*Int J Mol Sci* 2025)	Mechanism	Aberrant stretch activation/pulsatile myofilament activity as tremor driver
37070061 (Vial 2023)	Comparative / correction	Explicitly states myogenic tremor is classically an *MYBPC1* entity; reports a *distinct* first-of-its-kind *MYH7* case — anchors the gene correction
23153285	Comparative biology	Natural *MYH7* tremor-myopathy (porcine Campus syndrome) — different gene, analogous phenotype

Note on the gene correction. Early in this investigation the disease was provisionally attributed to *MYH7* based on the "myogenic tremor" literature. Authoritative identifier mapping (MONDO:0032797 = OMIM #618524 = CMYO16 = MedGen C5231401) and the explicit statement in PMID: 37070061 — *"The concept of a myopathy with associated tremor ('myogenic tremor') in humans has been previously described for specific MYBPC1… variants. Here we report for the first time an individual with tremor who was found to have a de-novo likely pathogenic variant in Myosin Heavy Chain 7 (MYH7)"* — confirm that **CMYO16 is an *MYBPC1* disease.** The report is curated accordingly; *MYH7* content is retained only as clearly-labeled comparative material.

Limitations and Knowledge Gaps

1. **Ultra-rare disease:** Only ~16 patients/~8 families reported; all epidemiologic, penetrance, expressivity, and prognostic statements rest on very small numbers.
2. **No prevalence/incidence estimates** per 100,000; no registry.
3. **Limited variant spectrum:** Only four M-motif missense variants are established; genotype–phenotype correlations are preliminary.
4. **No omics signatures** (transcriptomic/proteomic/metabolomic) specific to CMYO16.
5. **Tremor mechanism partly inferred:** The sarcomere-origin model is strongly supported but the precise contribution of central (spinal-loop) amplification in humans is not fully resolved; key electrophysiology quotes come from an analogous *MYH7* case rather than a *MYBPC1* patient.
6. **No targeted therapy or clinical trials;** management is supportive and largely empirical.
7. **Histopathology** of CMYO16 is not uniformly codified.
8. **Biopsy/QoL data** are sparse; no validated disease-specific QoL instrument.

Proposed Follow-up Experiments / Actions

1. **Expand the patient registry:** Aggregate all reported *MYBPC1* M-motif cases (GeneMatcher, international networks) to refine penetrance, expressivity, natural history, and the frequency of neonatal respiratory involvement.
2. **Direct tremor electrophysiology in *MYBPC1* patients:** Perform accelerometry + multi-level EMG coherence studies specifically in CMYO16 patients to confirm sarcomere-origin and quantify any central-loop amplification (replicating the *MYH7*-case paradigm).
3. **Model additional variants:** Generate knock-in mice/iPSC-myotubes for L263R and Y247H to compare increased- vs decreased-binding mechanisms side by side with E248K.
4. **Myofilament mechanics:** Systematic ex-vivo stretch-activation assays across variants to define the biophysical signature that produces oscillation.
5. **Structural studies:** Cryo-EM/modeling of the sMyBP-C M-motif–myosin interface to explain how opposite binding changes converge on tremorgenesis.
6. **Therapeutic screening:** Use E248K-KI mice and patient myotubes to test sarcomere modulators (myosin inhibitors/activators) for tremor suppression.

7. **Biomarker/omics profiling:** Muscle transcriptomics/proteomics from patient biopsies to identify downstream signatures and candidate modifiers.
8. **Standardize diagnostic criteria** and biopsy interpretation for CMYO16 to aid differentiation from central tremor and other congenital myopathies.

Report compiled from OMIM/MedGen/HPO annotations and primary literature. Causal gene: MYBPC1 (sMyBP-C), chromosome 12q23.2. Disease: Congenital Myopathy-16 (CMYO16 / "Myotrem"), OMIM #618524, MONDO:0032797.
